Robust Monolithic 3D Carbon-Based Computing-in-SRAM With Variation-Aware Bit-Wise Data-Mapping for High-Performance and Integration Density

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***Abstract*— Bit-serial computing-in memory with SRAM cells (SRAM-CIM) enables a full set of integer and floating-point arithmetic operations and various data-intensive computations. Carbon nanotube field-effect transistors (CN-MOSFETs) with high scalability, energy-efficiency, and low process thermal budget are attractive to realize high-dense monolithic three-dimensional (M3D) SRAM-CIM. However, CN-MOSFETs possess unique process variations with asymmetric spatial correlations which can significantly influence the performance and reliability of carbon-based SRAM-CIM. In this paper, new M3D-4N4P SRAM-CIM cells with CN-MOSFETs are proposed with optimized profiles for achieving ultra-high integration density while preserving robustness of data-access and computation. Furthermore, the variation-aware bit-wise data-mapping method is proposed for enhancing the performance of carbon-based SRAM-CIM by leveraging the spatial correlations of CN-MOSFETs. By minimizing the area skew of vertically-stacked layers, the areas of proposed M3D-4N4P SRAM-CIM cells are reduced by up to 50.32% compared to the previous 6N2P SRAM-CIM cells assuming carbon nanotube transistor technology. The proposed M3D-4N4P SRAM-CIM array also achieves by up to 2.17× higher throughput on arithmetic operations and 18.34% lower computing latency with 25.36% reduced energy consumptions on MAC-based benchmarks, respectively, compared to the previous 2D-6N2P SRAM-CIM array.**

***Index Terms*— Carbon nanotube transistor, SRAM, bit-serial in-memory computing, monolithic 3D, variation-aware bit-wise data-mapping.**

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I. Introduction

SRAM-BASED in-memory computing has been widely explored for varieties of data-intensive applications, such as general purpose computing [1], [2], [3], [4], [5], [6], [7], application-specific accelerators aiming for neural networks with multiply-and-accumulate (MAC) operations [8], [9], [10], [11], [12], [13], and data encryption/decryption [14]. As software algorithms are evolved rapidly and demand for high computation flexibility, implementing multi-precision integer and floating-point arithmetic operations in SRAM-CIM is attractive for prevailing broad-purpose computations.

Among the state-of-art CIM techniques implemented with high-performance and maturity SRAM, the traditional silicon-based bit-serial SRAM-CIM [4], [5], [6] enables a full set of integer and floating-point arithmetic operations with wide range of bit-widths by serially processing each bit of operands. To maintain compatibility with mainstream processor architecture, two extra n-channel MOSFETs are added to the standard 6-transistor (6T) SRAM cell to form a 6-n-channel and 2-p-channel (6N2P) transposable bit-serial 8T SRAM-CIM cell [4] in conventional 2D integration. Consequently, the integration density of the previously published silicon-based transposable bit-serial 6N2P SRAM-CIM array [4] is inevitably degraded compared to the standard 6T SRAM array.

Monolithic three-dimensional (M3D) technology can be employed to enhance the integration density of SRAM-CIM arrays with gate-level [15], [16], [17], [18] and transistor-level [25] partitions. With the conventional silicon transistors, the standard 6T SRAM cell and extra assisted transistors are placed into two different layers with FinFET-based gate-level M3D technology to form various in-memory operations such as comparator and XOR [15], [16], [17], [18]. Dedicated low-thermal-budget laser process [15], [16], [17], [18] is required to enable the silicon-based M3D while alleviating the problems of degraded performance in silicon devices of upper layers induced by high-temperature processing. The previous M3D SRAM-CIM designs [15], [16], [17], [18] also exhibit sub-optimal placement of transistors, which can induce significant area skews between different layers. Besides, the silicon CMOS is stressed by physical-scaling limitations in deep-nanometer technology node with increased leakage currents

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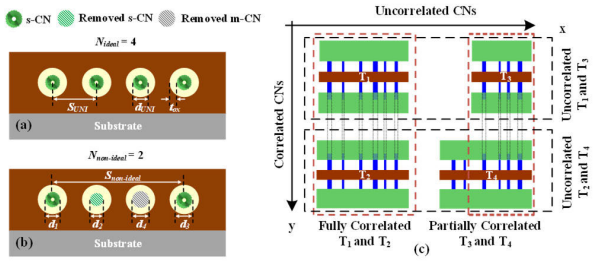


Fig. 1. GAA-CN-MOSFET and spatial correlations of multiple transistors [25]. (a) Top-view of an ideal GAA-CN-MOSFET with nominal diameter of d_{UNI} and nominal pitch of s_{UNI}. The ideal number of tubes (N_{ideal}) is 4. (b) A non-ideal GAA-CN-MOSFET after removing the m-CNs induced by diameter variations. The non-ideal number of tubes ($N_{non-ideal}$) is reduced to 2. (c) The spatial correlations of multiple CN-MOSFETs along the growth direction of CNs.

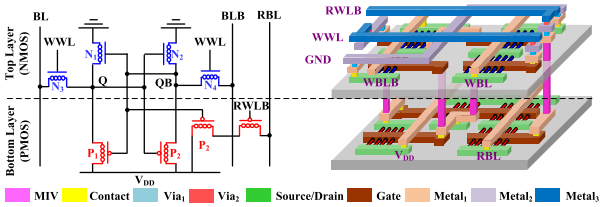


Fig. 2. Previously published M3D-4N4P SRAM cell (without computing-in-memory functionality) achieving balanced area between different layers [25].

[19], [20], [21], exacerbating the thermal issues in M3D. Therefore, it still remains a great challenge to achieve high-performance M3D SRAM-CIM while preserving integration density with conventional silicon transistors.

Carbon nanotube MOSFETs (CN-MOSFETs) are attractive alternatives to silicon for M3D circuit design [22], [23], [24], [25], [26], [27], enabled by their naturally low fabrication temperature. The well suppressed subthreshold leakage currents of deep-nanoscale CN-MOSFETs [28], [29], [30], [31], [32] also alleviate the thermal budget of M3D. An ideal gate-all-around carbon nanotube MOSFET (GAA-CN-MOSFET) is shown in Fig. 1(a), where the channel is composed of pure semiconducting CNs (s-CNs) with nominal diameter d_{UNI} [25]. To achieve ultra-high density, the transistor-level M3D SRAM circuits based on GAA-CN-MOSFETs are presented in [25], with equal number of n-channel and p-channel transistors (4N4P) split into two different device layers connected by dense monolithic inter-layer vias (MIVs), as shown in Fig. 2. Since the same-sized n-channel and p-channel CN-MOSFETs have similar driving strengths (with equally high carrier mobilities) [30], the sizes of CN-MOSFETs are balanced in both device layers in the previous M3D-4N4P SRAM cell while preserving high performance for data-access without the need to upsize p-channel CN-MOSFETs. Besides, with the low transfer temperature of CNs and identical process steps in each layer [23], the performance of devices across different layers is not degraded in carbon-based M3D.

Although high integration density across multiple layers is achieved, currently reported carbon-based M3D memory circuits [25], [27] have no in-memory computing capability. Directly transferring the previous 2D-6N2P SRAM-CIM cell [4] into transistor-level M3D would induce area skew with significantly unbalanced number of n-channel (with number of 6) and p-channel (with number of only 2) devices in different layers. Besides, carbon-based M3D SRAM-CIM suffers from non-idealities related to carbon nanotubes. Firstly, despite their rapid maturity and significantly reduced defects [33], the carbon nanotube transistors are still subject to the specific device variations such as diameter variations, leading to the formation of metallic-CNs (m-CNs) which results in the short-circuit failure, as shown in Fig. 1(b). Though m-CNs can be efficiently removed in both layers of M3D circuits [22], reducing the number of tubes in transistors can lead to performance fluctuations in carbon-based circuits. Besides, permanent open-circuit may also occur if all the CNs are removed from the channel [25]. Secondly, transistor sizing for carbon-based CMOS with symmetric electrical driving strengths is needed for achieving high-performance and robust SRAM-CIM. Thirdly, carbon nanotube transistors exhibit asymmetric spatial correlations [34] in CIM array, as shown in Fig. 1(c). When implementing complex arithmetic operations with bit-serial algorithm across different memory columns, multiple CN-MOSFETs fabricated along the same CN-growth direction display spatial correlations with similar electrical properties. Consequently, the storage locations of operands and intermediate results directly influence the latency of carbon-based bit-serial SRAM-CIM circuits. Therefore, dedicated design for carbon-based high-performance and robust M3D SRAM-CIM circuits against the variations is pivotal when considering the symmetric CMOS electrical properties and spatial correlations of CN-MOSFETs.

In this work, two new bit-serial M3D-4N4P SRAM-CIM cells with carbon nanotube transistor technology are proposed for achieving ultra-high integration density with high performance and robustness in both SRAM and CIM modes. The profiles of the proposed M3D-4N4P SRAM-CIM cells are optimized by trading off different design metrics in SRAM and CIM modes, considering the symmetric CMOS electrical properties of CN-MOSFETs. A variation-aware bit-wise data-mapping (VBM) method is also proposed for enhancing the performance of carbon-based SRAM-CIM by leveraging the spatial correlations of CN-MOSFETs. The layout areas of proposed M3D-4N4P SRAM-CIM cells are reduced by 50.32% and 17.97% compared to the previously published bit-serial 2D-6N2P SRAM-CIM cells and its M3D version based on carbon nanotube transistors, respectively. By employing the variation-aware bit-wise data-mapping method, the proposed carbon-based M3D-4N4P SRAM-CIM array achieves by up to 2.17× higher throughput on arithmetic computing task and 18.34% lower computing latency with 25.36% reduced energy consumptions on MAC-based benchmarks, respectively, compared to the previous 2D-6N2P SRAM-CIM array.

The reminder of this paper is organized as follows. In Section II, operations, challenges, and optimizations of previous 6N2P SRAM-CIM cell are discussed. In Section III, ultra-high density M3D-4N4P SRAM-CIM cells are proposed. In Section IV, a novel mapping method is introduced for high-performance carbon SRAM-CIM array. Extensive simulations and discussions are presented in Section V. Finally, conclusions are given in Section VI.

II. Previously Published 6N2P SRAM-CIM Cell Based on Carbon Nanotube Transistor Technology

In this section, the previously published 2D-6N2P SRAM-CIM cell [4] based on CN-MOSFETs is reviewed. The

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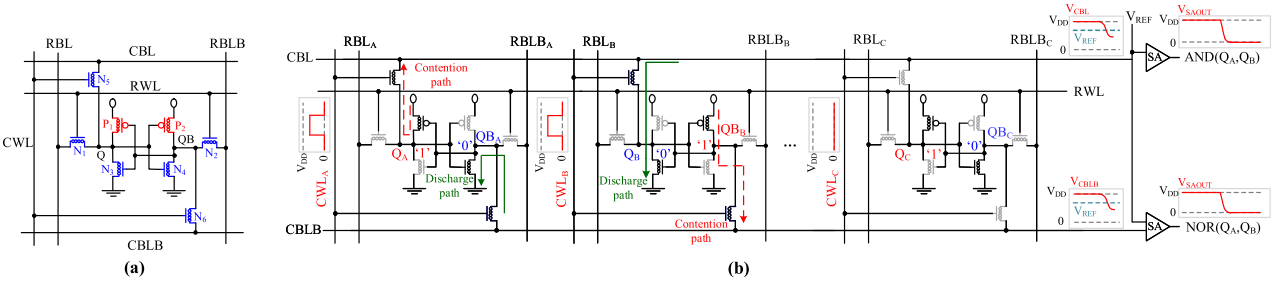


Fig. 3. Schematic and operations of 6N2P SRAM-CIM cell based on CN-MOSFETs. (a) Schematic. (b) In-memory computing for AND and NOR logic operations with datatype '10'. *RWL*: row wordline. *CWL*: column wordline. *RBL*/*RBLB*: bitlines for operations in SRAM mode. *CBL*/*CBLB*: bitlines for operations in CIM mode.

operations and challenges of 2D-6N2P SRAM-CIM cell based on CN-MOSFETs are described in Section II-A. In Section II-B, the previous 6N2P SRAM-CIM cells are optimized with profiles and layouts for 2D and M3D designs.

A. Operations and Challenges of 2D-6N2P SRAM-CIM Cell Based on Carbon Nanotube Transistor Technology

The schematic of 2D-6N2P SRAM-CIM [4] cell assuming carbon nanotube transistor technology is shown in Fig. 3(a). Compared with the standard 6T SRAM-CIM cell, 2D-6N2P SRAM-CIM cell adds additional transposable access transistors N_5 and N_6 for transposable data-access. 2D-6N2P SRAM-CIM cell is attractive for supporting bit-serial floating arithmetic directly in memory without the need to align data bits in columns [4], which is impractical in standard 6T SRAM based CIM works [1], [3] with unidirectional data-access.ss

The 2D-6N2P SRAM-CIM cell supports both SRAM and CIM modes. In SRAM mode, row-wise read and write operations are done as standard 6T SRAM by disabling N_5 and N_6. To initiate row-wise read operation, *RWL* transitions high. The pre-charged *RBL* and *RBLB* are discharged through path N_1-N_3 and N_2-N_4, respectively. Data read on *RBL* and *RBLB* is sensed by the double-ended sense amplifier (D-SA). The row-wise write operation is done by activating *RWL* and forcing new data from *RBL* and *RBLB* to the storage nodes Q and QB, respectively. The read and write margins of 2D-6N2P SRAM-CIM cell are positively related with the read-path ratio (N_3/N_1 or N_4/N_2) and write-path ratio (N_1/P_1 or N_2/P_2), respectively. In CIM mode, column-wise read, write, and logic operations are supported. The column-wise read and write operations are done in a transposed manner as SRAM mode by using the transposable access transistors N_5 and N_6 while disabling row-wise access transistors N_1 and N_2. For column-wise logic operations, operands are transposed for bit-wise logic computations by simultaneously activating two *CWLs* while *RWLs* maintain low, as shown in Fig. 3(b). The pre-charged *CBL* (*CBLB*) will be discharged whenever there is at least one pull-down path connected to *CBL* (*CBLB*). The discharged voltage produced on *CBL* (*CBLB*) is sensed by D-SA with reference voltage to output the logic result AND (NOR). During logic operations, potential contention paths may occur, slowing down the discharge process of bitlines when two 6N2P SRAM-CIM cells stores different logic states, and can further lead to runtime data corruption due to the dynamic read disturbance [35]. Nevertheless, the contention effect and read disturbance can be significantly mitigated by maintaining relatively high bitline voltages for reliable sensing, since the source voltages of access transistors are still high enough before the access transistors shifting from cutoff to linear region [35].

The previously published 2D-6N2P SRAM-CIM cells based on CN-MOSFETs still face several design challenges. Firstly, non-6T SRAM-CIM cell imposes a larger cell area. Employing M3D technology is urgently needed for achieving compact layout to minimize the impact of density loss in 2D-6N2P SRAM-CIM cell. Secondly, the robustness of data-access in both SRAM and CIM modes is strongly affected by the relative strengths of correlated CN-MOSFETs. Due to the process imperfections, the diameters and number of carbon nanotubes can be randomly changed as well as the strengths of CN-MOSFETs, which may incur degraded read and write margins for SRAM-CIM cells. Besides, the symmetric electrical driving strengths of carbon nanotube transistors also need to be considered. A proper sizing method of SRAM-CIM cells is needed for the robust data-access in both SRAM and CIM modes. Thirdly, due to the process imperfections, CN-MOSFETs suffer from performance variations with spatial correlations. Using low-performance SRAM-CIM cells (LPCs) instead of high-performance SRAM-CIM cells (HPCs) degrades the overall system throughput.

B. Optimizations of 6N2P SRAM-CIM Cell Based on CN-MOSFETs for 2D and M3D Technology

To determine the optimal size of 6N2P SRAM-CIM cell in carbon nanotube transistor technology for robust data-access, a modified quality metric (*QM*) considering both SRAM and CIM modes is developed as

$$QM = \frac{RSNM_{SRAM} \times CWLM_{SRAM}}{Delay_{SRAM}} \times \frac{RSNM_{CIM} \times CWLM_{CIM}}{Delay_{CIM}} \times \frac{Yield_{array}}{P_{leakage} \times Area_{cell}}, \times (Constraints : DR = 0\% and Yield_{array} \geq 95\%) \quad (1)$$

where $RSNM_{SRAM}$ and $CWLM_{SRAM}$ are the read static noise margin (read margin) and the combined word line margin (write margin) in SRAM mode, respectively, characterizing the robustness of SRAM operations. $Delay_{SRAM}$ is the delay of read operations in SRAM mode. The read delay is measured from the 50% transition of wordline (*RWL*) to 50% output transition of the D-SA. $RSNM_{CIM}$ and $CWLM_{CIM}$ are the

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TABLE I
THE OPTIMIZED NUMBER OF TUBES OF TRANSISTORS IN SRAM-CIM CELLS

Cell	P_1/P_2	P_3/P_4	N_1/N_2	N_3/N_4	N_5/N_6
2D/M3D-6N2P	6	N/A	5	7	5
M3D-4N4P CPRN	7	5	5	6	N/A
M3D-4N4P CNRP	6	5	5	7	N/A

read static noise margin and the combined word line margin in CIM mode, respectively. $Delay_{CIM}$ is the delay of logic operations in CIM mode. The CIM delay is measured from the 50% transition of wordline (*CWL*) to 50% output transition of the D-SA with reference voltage. Disturbance rate (*DR*) is the chance of data flip in logic operations between CIM cells. To ensure the reliability in logic operations, *DR* which is determined by at least 1000 Monte Carlo simulations should be limited to 0%. $Area_{cell}$ is the area of layout. $P_{leakage}$ is the leakage power of SRAM-CIM cell. $Yield_{array}$ is the functional yield of a 256×256-bit array which is calculated when all the transistors are functional. The simulations are done by using the Virtual Source GAA-CN-MOSFET model [37], assuming 16nm technology with 0.7V supply at 25 °C through the work. The number of carbon nanotubes of (6, 5, 7) are determined for (P_1 and P_2, N_1-N_2 and N_5-N_6, N_3 and N_4) as listed in Table I, because of the relatively high *QM* with small area and reliable read and wire margins satisfying the *DR* constraint.

The layout of 2D-6N2P SRAM-CIM cell is shown in Fig. 4(a). In the 2D layout, access transistors N_1(N_2) and N_5 (N_6) are fully correlated. Pull-down transistor N_3(N_4) and pull-up transistor P_1 (P_2) are partially correlated. Considering the spatial correlations, the functional yield of 256×256-bit 2D-6N2P SRAM-CIM array is

$$\begin{aligned} &P_{array_2D_6N2P} \\ &\quad = P_{cell_2D_6N2P}^{256} = (P_{correlated_P_1N_3}(6,7) \\ &\qquad \times P_{correlated_N_1N_5}(5,5) \times P_{correlated_N_2N_6}(5,5) \\ &\qquad \times P_{correlated_P_2N_4}(6,7))^{256} = 98.14\%, \end{aligned} \tag{2}$$

where $P_{correlated}$ is the probability that the correlated transistors are both functional. $P_{correlated}$ is the function of process parameters P_m (the probability of CNs being metallic in channel), P_s(the probability of CNs being semiconducting), P_{Rm} (the probability of removing metallic CNs), and P_{Rs}(the probability of wrongly removing the semiconducting CNs in channel when removing metallic CNs) [25]. P_m, P_s, P_{Rm}, and P_{Rs} are assumed to be 3%, 97%, 99.99%, and 2.5% [25], respectively, to evaluate the functional yield. A target yield larger than 95% is ensured.

To enhance the integration density, the previous 2D-6N2P SRAM-CIM cell based on CN-MOSFETs can be implemented with M3D technology where n-channel and p-channel transistors are split into different layers with 39.43% reduced area overhead compared to the 2D design. The optimized layout of M3D-6N2P SRAM-CIM cell is shown in Fig. 4(b). In the top layer, 6 n-channel transistors with 3 metal layers are tightly correlated. In the bottom layer, 2 p-channel transistors are connected to the top layer by 4 MIVs. Note that it is preferred to reserve the top and bottom layers for the n-channel and p-channel transistors, respectively, to reduce total metal layer

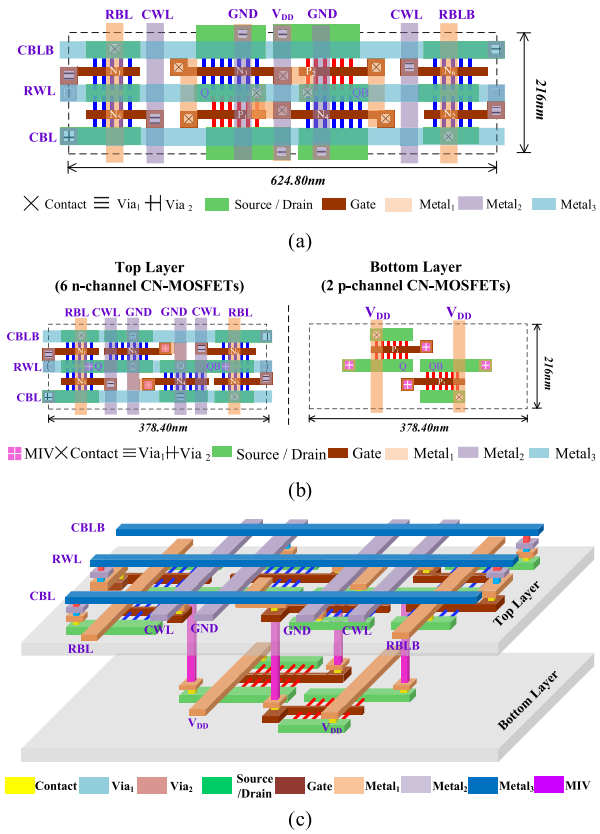


Fig. 4. Layout of 6N2P SRAM-CIM cell assuming carbon nanotube transistor technology. (a) Optimized 2D technology layout. The total area is 624.80nm × 216nm = 134956.80nm^2. (b) Optimized M3D technology layout. The total area is 378.40nm × 216nm = 81734.40nm^2. (c) Three dimensional view of layout.

consumptions while minimizing the heat dissipation and fabrication cost [36], as shown in the three dimensional view of M3D-6N2P SRAM-CIM cell in Fig. 4(c). The functional yield of 256×256-bit M3D-6N2P SRAM-CIM array is calculated as 97.25% which is slightly lower than 2D-6N2P SRAM-CIM array due to the less correlated layout. Besides, large area skew is induced by unbalanced transistor number between top and bottom layers in M3D-6N2P SRAM-CIM cell, leading to density loss. Mixing n-channel and p-channel devices in the same layer to reduce area skew inevitably increases the fabrication cost [36]. Therefore, a novel SRAM-CIM circuit is needed to better exploit the potential advantages of M3D technology for enhanced integration density while maintaining robust data-access and ensuring high-performance in-memory computing.

III. PROPOSED ULTRA-HIGH DENSITY M3D SRAM-CIM CELLS BASED ON CARBON NANOTUBE TRANSISTORS

In this section, two new carbon-based M3D-4N4P CPRN (column-wise operation with p-channel transistors and row-wise operation with n-channel transistors) and CNRP (column-wise operation with n-channel transistors and row-wise operation with p-channel transistors) SRAM-CIM cells are proposed for ultra-high integration density. Design details of proposed M3D-4N4P CPRN and CNRP SRAM-CIM cells are presented in Sections III-A and III-B, respectively.

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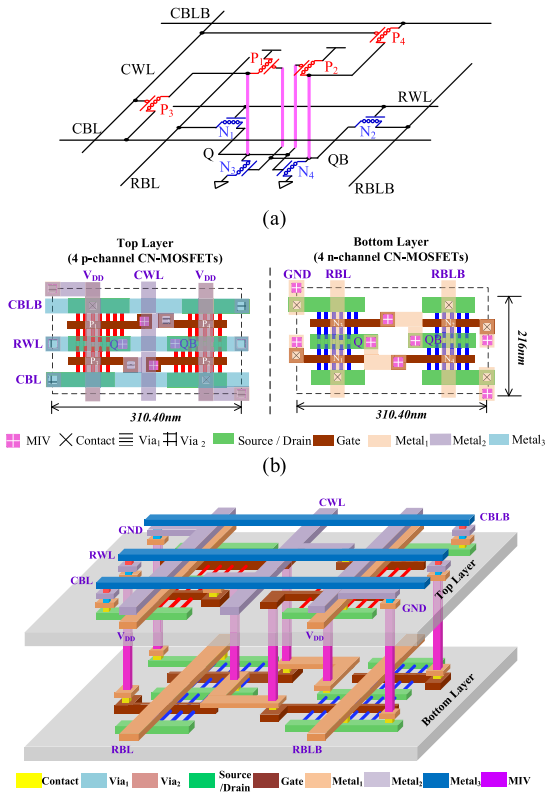


Fig. 5. Schematic and layout of proposed M3D-4N4P CPRN SRAM-CIM cell. The total area is 310.40nm × 216nm = 67046.40nm^2. (a) Schematic. (b) Layout with p-channel devices on top layer and n-channel devices on bottom layer. (c) Three dimensional view of optimized layout.

A. Proposed M3D-4N4P CPRN SRAM-CIM Cell Based on Carbon Nanotube Transistor Technology

The schematic of proposed M3D-4N4P CPRN SRAM-CIM cell is shown in Fig. 5(a). The proposed M3D-4N4P CPRN SRAM-CIM cell differs from the previously published 6N2P SRAM-CIM cell by replacing the n-channel row-wise access transistors N_5 and N_6 with the p-channel access transistors P_3 and P_4. This change results in an equal number of n-channel and p-channel transistors, achieving a balanced placement of transistors in both top and bottom device layers without affecting the performance of SRAM-CIM cells, thanks to the symmetric electrical properties of CN-MOSFETs.

The proposed M3D-4N4P CPRN SRAM-CIM cell supports both SRAM and CIM modes. In SRAM mode, row-wise read and write operations are supported, as shown in Figs. 6(a) and (b), respectively. Read operation is initiated by activating *RWL*. Pre-charged *RBLB* (*RBL*) discharges, while *RBL* (*RBLB*) maintains high when storage node QB (Q) stores '0'. The D-SA detects the differential voltages of *RBL* and *RBLB* to output data. Row-wise write operation is finished by activating *RWL* to turn on access transistors N_1 and N_2 after the intended data are put on *RBL* and *RBLB*, forcing the data into the storage nodes.

In CIM mode, column-wise read, write, and logic operations are supported in M3D-4N4P CPRN SRAM-CIM cell. Column-wise read operation is initiated by pre-discharging *CBL* and *CBLB* to GND. *CWL* transitions low to turn on P_3 and P_4. The D-SA with reference voltage is employed to sense the voltage

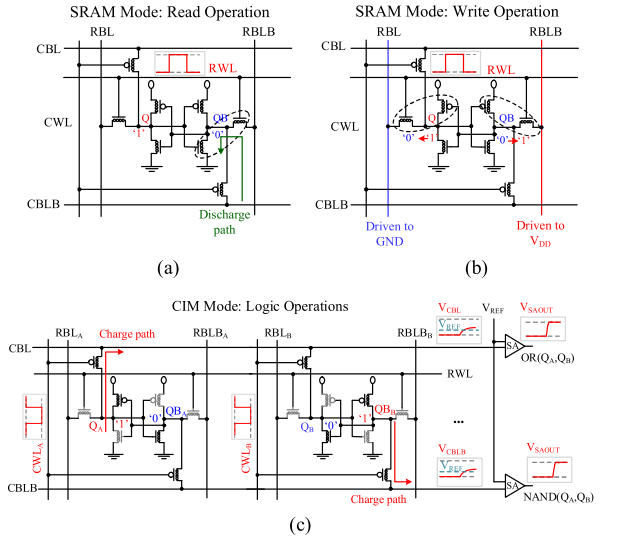


Fig. 6. SRAM and CIM modes of proposed M3D-4N4P CPRN SRAM-CIM cells. (a) Read operation in SRAM mode. (b) Write operation in SRAM mode. (c) Logic operation between two bits in CIM mode.

of *CBL* to output the data. It is worth noting that there is no need to upsize P_3 and P_4 to enhance the performance due to the symmetric driving strengths of carbon-based CMOS. To perform column-wise write operations, *CWL* transitions low to turn on P_3 and P_4. The intended data are forced into the storage nodes of M3D-4N4P CPRN SRAM-CIM cell through *CBL* and *CBLB*. The logic operation is done by activating two *CWL*s at the same time after pre-discharging *CBL* and *CBLB* to GND. *CBL* and *CBLB* are then conditionally charged depending on the states of two operands. When Q_A and Q_B store different logic states, such as (1,0) (or (0,1)), as shown in Fig. 6(c), both *CBL* and *CBLB* are charged. When Q_A and Q_B store the same logic states, such as (1,1) (or (0,0)), *CBL* (*CBLB*) is charged while *CBLB* (*CBL*) maintains low. By using D-SAs with reference voltage connected to *CBL* and *CBLB*, Boolean logic results of OR and NAND are obtained, respectively.

For the proposed M3D-4N4P CPRN SRAM-CIM cell, the read and write margins in SRAM mode are positively related to the read-path ratio (N_3/N_1 or N_4/N_2) and write-path ratio (N_1/P_1 or N_2/P_2), respectively. Therefore, large sizes of N_3 and N_4 but small sizes of P_1 and P_2 are preferred to achieve reliable read and write operations in SRAM mode. However, in CIM mode, the read and write margins are positively related to the coupled read-path ratio (P_1/P_3 or P_2/P_4) and write-path ratio (P_3/N_3 or P_4/N_4), respectively. The large sizes of P_1 and P_2 but small sizes of N_3 and N_4 are preferred for high-performance and reliable logic operations. Therefore, there are conflict sizing requirements between SRAM and CIM modes in the proposed M3D-4N4P SRAM-CIM cell.

To ensure reliable operations in both SRAM and CIM modes, the sizing rule using *QM* method proposed in Section II-B is applied. The number of carbon nanotubes of (7, 5, 5, 6) are determined for (P_1 and P_2,P_3 and P_4, N_1 and N_2, N_3 and N_4), respectively, to achieve robust SRAM and CIM modes, as listed in Table I. The profiles of proposed M3D-4N4P SRAM-CIM cell are optimized specifically to avoid dynamic read disturbance. The read disturbance is prone to occur when the write-path ratio is large, where the data on

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bitlines can be easier to overwrite the original data stored in SRAM-CIM cells involved in logic operations. To avoid the unintended data flip in CIM, the write-path ratio of proposed M3D-4N4P CPRN SRAM-CIM cell is thereby suppressed to be less than 1, while remaining sufficiently high for normal writing. The CNs are properly aligned in different transistors of proposed M3D-4N4P CPRN SRAM-CIM cell to achieve compact layout. As shown in Fig. 5(b), the access transistor $N_1(N_2)$ and pull-down transistor N_3 (N_4) are partially correlated, as well as the pull-up transistor $P_1(P_2)$ and access transistor P_3 (P_4). Besides, the p-channel and n-channel CN-MOSFETs are preferred to be placed into top layer and bottom layer, respectively, to reduce the metal layer consumptions for low fabrication cost and less heat dissipation, as shown in the three dimensional view of M3D-4N4P CPRN SRAM-CIM cell in Fig. 5(c). To reduce the number of MIVs with less parasitics, the MIVs of GND and *RWL* in the proposed M3D-4N4P CPRN SRAM-CIM cell are placed on the layout boundaries as shown in Fig. 5(b). In each CIM cell, the MIV of GND is shared by four surrounding cells, while the MIV of *RWL* is shared between two horizontally neighboring cells. Therefore, the proposed M3D-4N4P CPRN SRAM-CIM cell only consumes 1.5 more MIVs than the previous M3D-6N2P SRAM-CIM cell on average.

Considering the spatial correlations, the functional yield of proposed 256 × 256-bit M3D-4N4P CPRN SRAM-CIM array is evaluated as 97.98%. A target yield larger than 95% is ensured. The area of proposed M3D-4N4P CPRN SRAM-CIM cell is estimated to be 67046.40nm^2. The area overhead is thereby reduced by 50.32% and 17.97% compared to the previous 2D and M3D-6N2P SRAM-CIM cells, respectively.

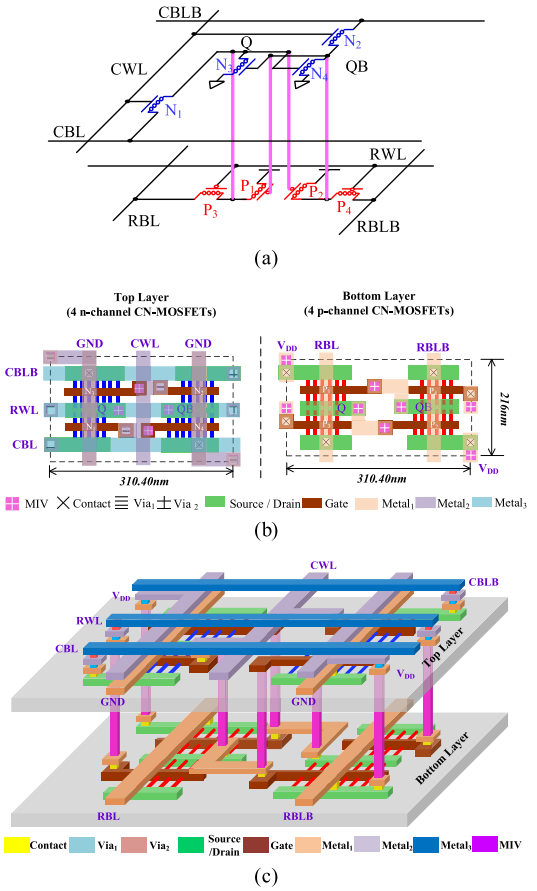


Fig. 7. Schematic and layout of proposed M3D-4N4P CNRP SRAM-CIM cell. The total area is 310.40nm × 216nm = 67046.40nm^2. (a) Schematic. (b) Layout with n-channel devices on top layer and p-channel devices on bottom layer. (c) Three dimensional view of optimized layout.

B. Proposed M3D-4N4P CNRP SRAM-CIM Cell Based on Carbon Nanotube Transistor Technology

The schematic of proposed M3D-4N4P CNRP SRAM-CIM cell is shown in Fig. 7(a). The access transistors N_1 and N_2 in 6N2P SRAM-CIM cell are replaced by P_3 and P_4 in M3D-4N4P CNRP SRAM-CIM cell. The replacement of access transistors induces unconventional row-wise read and write operations in SRAM mode which are shown in Figs. 8(a) and (b), respectively. To initiate row-wise read operation in SRAM mode for M3D-4N4P CNRP SRAM-CIM cell, *RBL* and *RBLB* are pre-discharged to GND. *RWL* then transitions low to activate P_3 and P_4. The D-SA is employed to sense the voltage difference of *RBL* and *RBLB* to output data. Row-wise write operation in SRAM mode is done by setting *RWL* low to overwrite the state of storage nodes Q and QB after data are placed onto *RBL* and *RBLB*. The operations in CIM mode of M3D-4N4P CNRP SRAM-CIM cell are similar to 6N2P SRAM-CIM cell which is shown in Fig. 8(c), thus detailed descriptions are omitted here for simplicity.

The layout of M3D-4N4P CNRP SRAM-CIM cell is shown in Fig. 7(b). The n-channel and p-channel devices are preferred to be placed into top and bottom layers, respectively, to reduce metal layer consumptions. Using *QM* based sizing method, the number of carbon nanotubes of pull-up transistors P_1 and P_2, pull-down transistors N_3 and N_4, p-channel access transistors P_3 and P_4, and n-channel access transistors N_1 and N_2 are tuned to be 6, 7, 5, and 5, respectively, as listed in Table I. Similar to the analysis in the Section III-A, the write-path ratio of proposed M3D-4N4P CNRP SRAM-CIM cell is optimized to be less than 1 to avoid read disturbance in logic operations. Besides, the MIVs of V_{DD} and *RWL* in each cell are placed on the layout boundaries, which are shared by the neighboring cells to reduce the average number of MIVs, as shown in Fig. 7(b). The three dimensional view of M3D-4N4P CNRP SRAM-CIM cell is also shown in Fig. 7(c).

Considering the spatial correlations, the functional yield of proposed 256 × 256-bit M3D-4N4P CNRP SRAM-CIM array is estimated as 97.98%. The area of M3D-4N4P CNRP SRAM-CIM cell is estimated to be 67046.40nm^2 which is the same as M3D-4N4P CPRN SRAM-CIM cell. The area overhead is thereby reduced by 50.32% and 17.97% compared to the previous 2D and M3D-6N2P SRAM-CIM cells, respectively.

IV. Proposed Variation-Ware Bit-Wise Data-Mapping Method for 8T Transposable SRAM-CIM Array Based on Carbon Nanotube Transistor Technology

The carbon-based SRAM-CIM arrays suffer from the intrinsic process imperfections of CN-MOSFETs as discussed in Section I. The overall architecture of 256 × 256-bit 8T transposable SRAM-CIM array based on CN-MOSFETs is shown in Fig. 9(a). The near-memory computing (NMC) unit is connected with *CBL* and *CBLB* in each row to implement

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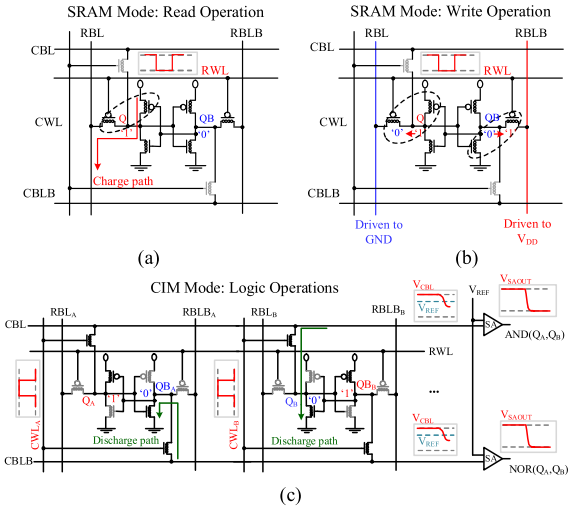


Fig. 8. SRAM and CIM modes of proposed M3D-4N4P CNRP SRAM-CIM cells. (a) Read operation in SRAM mode. (b) Write operation in SRAM mode. (c) Logic operation between two bits in CIM mode.

the bit-serial arithmetic operations. The conventional bit-serial CIM algorithm directly maps the bits of data into consecutive column addresses with bit-sequence order, inevitably using LPCs induced by the process variations for computing, which results in degraded performance of CIM array as described in Section II-A. To efficiently map the data (including operands, intermediate, and final results) involved in the single or multiple computation tasks with enhanced CIM performance, the variation-aware bit-wise data-mapping (VBM) method is proposed by leveraging the intrinsic spatial correlation characteristics of CN-MOSFETs. The VBM method is introduced in Section IV-A. The examples of VBM method for single computation tasks (arithmetic operations) and multiple computation tasks (MAC operations) are introduced in Section IV-B and Section IV-C, respectively.

A. Proposed Variation-Aware Bit-Wise Data-Mapping Method

The proposed VBM method uses the inherently high-performance SRAM-CIM cells in CIM array for latency critical operations to enhance CIM performance regardless of process variations in CN-MOSFETs. To support VBM method, PCA-LUT is used to store the transferred column addresses of SRAM-CIM array which are indexed by the original bit-sequence-ordered column addresses, as shown in Fig. 9(a). According to PCA-LUT, the data involved in latency critical operations can be mapped into the HPCs randomly distributed in CIM array. The PCA-LUT is employed for VBM method through the two-step data loading process for k q-bit operands, as shown in Fig. 9(b), where the loading columns are the last q columns of the CIM array. In the first step, the operands are loaded using row-wise write operations in SRAM mode for k cycles. In the second step, the word-wise parallelly loaded data are remapped into CIM array according to PCA-LUT in q cycles for each bit. To read out the final results in the remapped CIM array, the above two steps are reversed, where the bits out- of-order are firstly rearranged into the loading columns in bit-sequence order and read out by row-wise read operations.

Algorithm 1 Variation-Aware Bit-Wise Data-Mapping Method for a Multi-bit Data in Single Computation Task

Input: Data A[q-1:0] to be mapped to the q columns of the $R \times C$-bit array in the rth row
Output: PCA-LUT [q-1:0]
Precondition: the number of unoccupied columns in the rth row is sufficient for a certain single computation task
0: Evaluate latency rank vector LRV[C-1:0] and occupation vector OV[C-1:0] of C columns from the longest latency to the shortest latency
Initialize PCA-LUT[q-1:0] to zeros
1: **for** (i =0 to q-1) **do** // make judgement for each bit
2: **if** (A[i] is involved in read or CIM operations) **then**
3: **for** (j = 0 to C-1) **do** // fastest to slowest column
4: **if** (OV[j] is 0) **then** // LRV[j] is available
5: A[i] is mapped into the LRV[j]th column
6: Update OV[j] to 1
7: PCA-LUT[i]= LRV[j]
8: Break
9: **end if**
10: **end for**
11: **else** // A[i] is only involved in write operations
12: **for** (j = C-1 to 0) **do** // slowest to fastest column
13: **if** (OV[j] is 0) **then** // LRV[j] is available
14: A[i] is mapped into the LRV[j]th column
15: Update OV[j] to 1
16: PCA-LUT[i]= LRV[j]
17: Break
18: **end if**
19: **end for**
20: **end if**
21: **end for**

The VBM method for a multi-bit data in single computation task is shown in Algorithm 1. The data A[q-1:0] involved in a single computation task (such as operands, intermediate, and final results) is used as an example to demonstrate Algorithm 1. To conduct bit-serial computing, the bits of data A[q-1:0] are mapped into the q columns of rth row in the $R \times C$-bit SRAM-CIM array. Before mapping, there is a precondition that the unoccupied memory columns are sufficient for mapping all the data bits involved in the single computation task. If there are not enough unoccupied memory columns to accommodate the data in the current row, the data will be mapped to another row with sufficient memory columns by repetitively searching and updating the row number r until the row with available memory columns is found. In the beginning of mapping, latency rank vector (LRV) and occupation vector (OV) of C columns in the rth row are initialized. LRV records the performance of CIM array, which contains C column addresses indexed by the performance rank. For example, the address of the highest performance column is stored in the first element of LRV. The performance rank is initialized with column-wise read latency detection before computations. Due to the spatial correlations, SRAM-CIM cells in the same column but different rows exhibit the similar performance, thereby the R rows in array share the same LRV. OV records the remaining available columns in a

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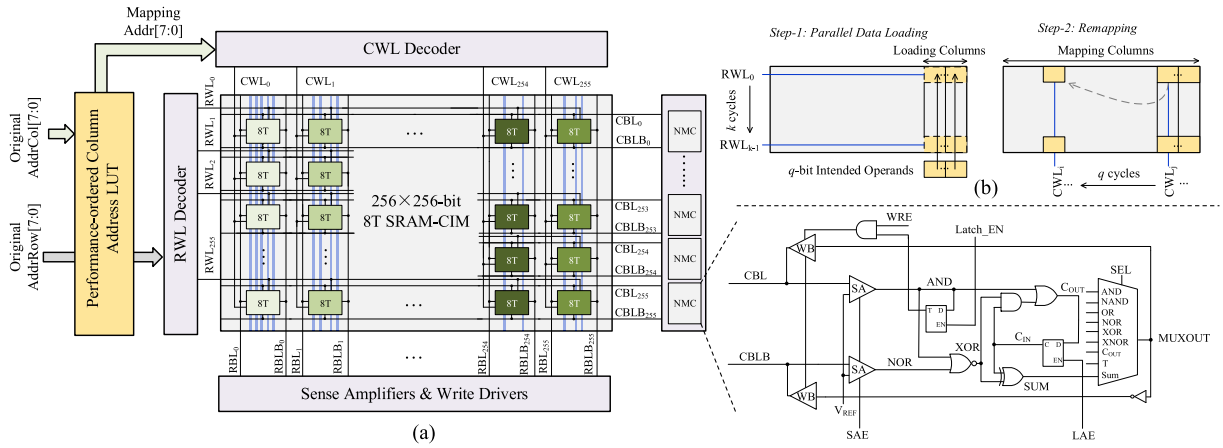


Fig. 9. Architecture of 8T transposable SRAM-CIM array supporting VBM method. (a) 256 × 256-bit 8T SRAM-CIM array with PCA-LUT. (b) Data loading scheme for CIM.

certain row for data mapping, and each row has its own OV for updating.

In the mapping process, the column address of each bit of data A is mapped to the performance-ordered memory columns according to operations of data bits involved in the CIM flow. If the ith bit of data A is involved in read or CIM operations (the latency critical operations), A[i] is mapped to the fast and unoccupied memory columns (unoccupied HPCs) according to LRV and OV. If the ith bit of data A, such as the final result bit, is only involved in write operations (the latency non-critical operations), then A[i] is mapped to the slow and unoccupied memory columns (unoccupied LPCs) according to LRV and OV. Once the mapping address of intended data bit is determined, the corresponding transferred address is stored into PCA-LUT. After mapping each bit, OV will be updated. By using Algorithm 1, data involved in a single computation task (fixed-point or floating-point arithmetic operation) can be stored and computed with bit-serial algorithm in high clock frequency regardless of CN-MOSFET variations.

The VBM method for vectors with n elements quantized as q bits in multiple computation tasks is shown in Algorithm 2. The length of vectors for multiple computation tasks usually exceeds the length of memory columns. To map the long vectors into the memory array, Algorithm 2 implemented in two steps is proposed without losing generality. In the first step, the storage constraints are determined. C_{temp} is evaluated to reserve memory columns for temporary results. C_{op} is calculated to generate the number of memory columns in each row for storing the operands for computations. To check if current row is available, C_{occ} is used to represent the number of columns occupied by data in the top fast C_{op} columns of current row. In the second step, the row which satisfies the storage constraints will be selected to store vectors. After mapping, C_{occ} will be updated. When the row is not available, the next row will be judged to store the vectors until whole vectors are mapped into the array.

Algorithm 2 Variation-Aware Bit-Wise Data-Mapping Method for Vectors in Multiple Computation Tasks

Input: Vectors A[n-1:0][q-1:0] and B[n-1:0][q-1:0] to be mapped to the $R \times C$-bit array

Output: Mapping of Vectors A[n-1:0][q-1:0] and B[n-1:0][q-1:0]

0: Initialize parameters:
Reserve columns C_{temp}// Take MAC for an example, C_{temp} = ⌈2×q+ $\log_2(n)$⌉, where $\log_2(n)$ bits are used for accumulation extension
C_{op} = ⌈C/2-2×C_{temp}⌉ // operands space
Select the intended r^{th} row to store operands
Number of columns C_{occ} occupied by operands in the top C_{op} columns of rth row
1: **for** (i =0 to n-1) **do** // for every element in the vector
2: **if** (C_{occ}+2×q ≤ C_{op}) **then**
3: store A[i][q-1:0] and B[i][q-1:0] to top 2×q columns of rth row according to **Algorithm 1**
4: C_{occ} + =2×q // update occupation
5: **else do** // r^{th} row is full, select the next row
6: **while** (C_{occ}+2×q > C_{op})
7: r = r+1 // Update the r value
8: Update C_{occ} according to the rth row
9: store A[i][q-1:0] and B[i][q-1:0] to top 2× q columns of rth row according to **Algorithm 1**
10: **end if**
11: **end for**

B. Mapping Processes of Single Computation Tasks (Arithmetic Operations) With Proposed VBM Method

In this section, the arithmetic operations are evaluated as examples of single computation tasks. For bit-serial fixed-point addition, each bit of two operands stored in the memory columns of certain row does bit-wise addition to serially generate final sum [4]. So each bit of two operands are involved in CIM operations, while each bit of sum is only involved in write operation. A 2-bit addition is serially processed in 3 cycles as shown in Fig. 10(a). In the first 2 cycles, *CWLs* of operands are activated to perform single bit addition with writing back the sum. The carryout of each cycle are stored in the latch of NMC. The final cycle is used to write back carryout. For the case without VBM method, LPCs may be used in logic operations which incurs large latency. However, when only using HPCs for CIM and LPCs for writing final results, the overall latency of single cycle is shortened, as shown in

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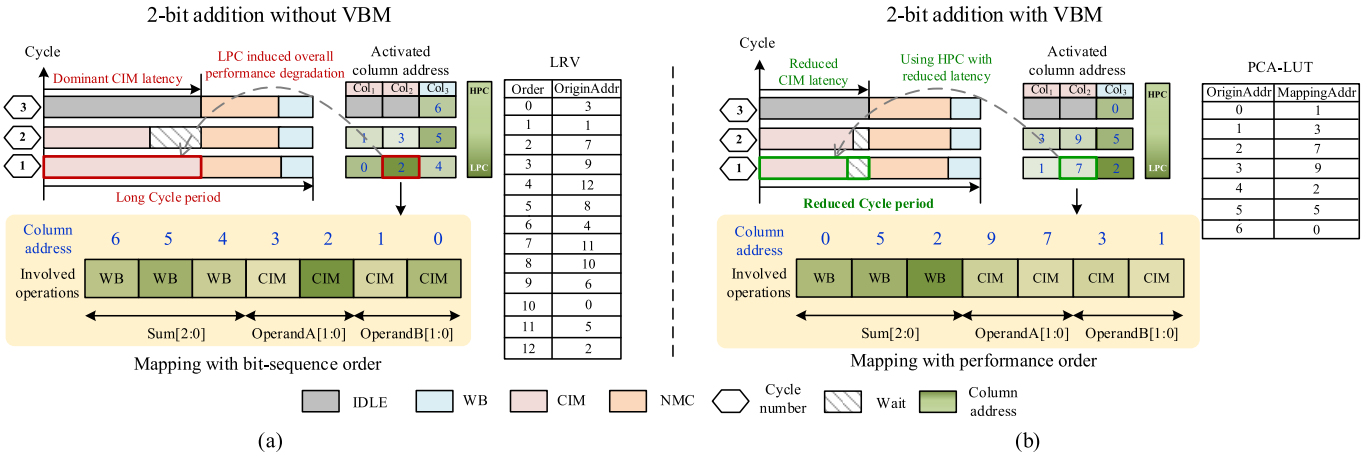


Fig. 10. Illustration of the proposed VBM method using 2-bit addition as an example. The block length presents the latency of operations. The shade of green color presents the performance of SRAM-CIM cells where the lower performance cell is in the darker color. IDLE: the CIM cell is idle without any operations. WB: write back. CIM: computing-in-memory for logic operation. NMC: near-memory computing. (a) 2-bit addition without mapping method. (b) 2-bit addition with mapping method, and the cycle period is reduced as compared to (a).

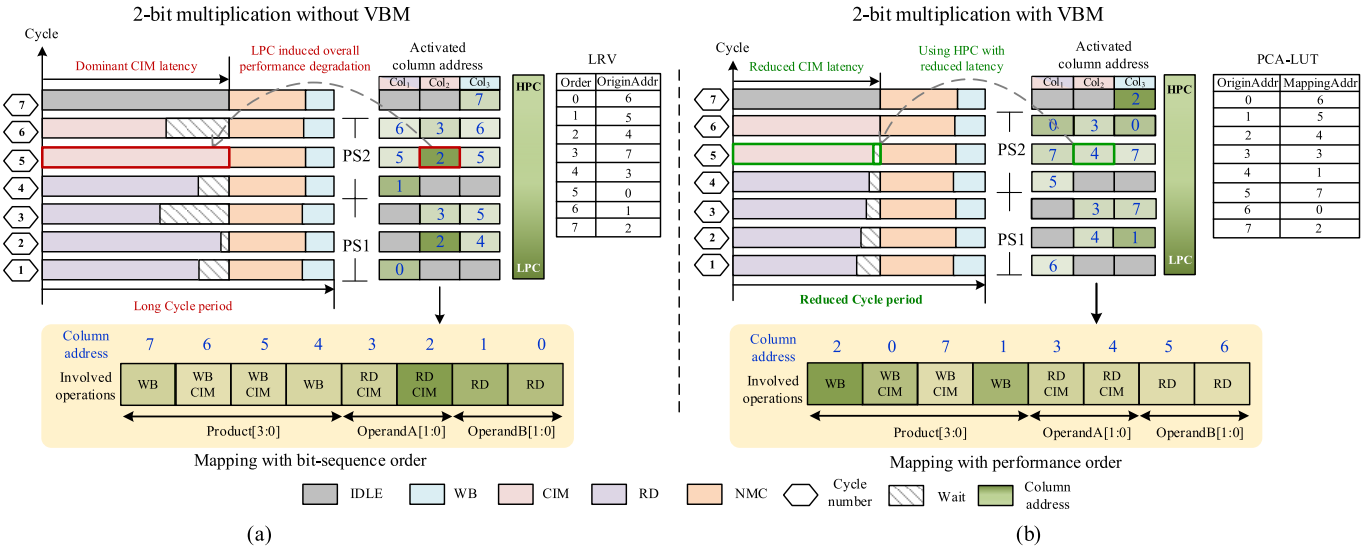


Fig. 11. Illustration of the proposed VBM method using 2-bit multiplication as an example. IDLE: the CIM cell is idle without any operations. WB: write back. CIM: computing-in-memory for logic operation. RD: read operation. NMC: near-memory computing. (a) 2-bit multiplication without mapping method. (b) 2-bit multiplication with mapping method, and the cycle period is reduced as compared to (a).

Fig. 10(b). Using Algorithm 1, each bit of operands and sum can be correctly mapped into memory columns, where the bit cells used in CIM operations are all mapped into the light-green blocks which present HPCs, thus a reduced clock period is expected as compared to the case without mapping method.

For bit-serial fixed-point multiplication, the multiplicand, multiplier, and product need to be mapped into memory columns. During multiplication, each bit of multiplier is read out and loaded to the latch of NMC. When the first bit of multiplier is loaded into latch, multiplicand is conditionally copied to generate partial sum depending on the value stored in latch. For the rest bits of multiplier, bit-wise addition is conditionally conducted between multiplicand and partial sum until final sum are generated in the product columns [4].

In the above process, each bit of multiplier is involved in read operations, and each bit of multiplicand is involved in CIM operations. The least significant bit (LSB) and the most significant bit (MSB) of product are only involved in write operations while the rest bits of product are involved in write and CIM operations. As shown in Fig. 11(a), a 2-bit multiplication is finished in 7 cycles with read, CIM and write operations. When read or CIM operations happens in the low-performance bit cell like the 2nd column in cycle 5, long latency is incurred which enlarges the clock period and degrades the system performance. While using the proposed VBM method, as shown in Fig. 11(b), the overall computing latency is shortened by using HPCs for read and CIM operations.

For bit-serial fixed-point subtraction, additional temp results are needed to store the inverted subtrahend. Then temp results are added with minuend. So each bit of minuend, subtrahend, temp results, and difference results are involved in CIM, read, CIM, and write operations, respectively. For bit-serial fixed-point division, each bit of dividend is sequentially copied to remainder, and remainder is subtracted with divisor to generate quotient. Temp columns are used to record subtraction results and update remainder. Therefore, divisor, dividend, quotient, remainder, and temp results are involved in CIM, read, write,

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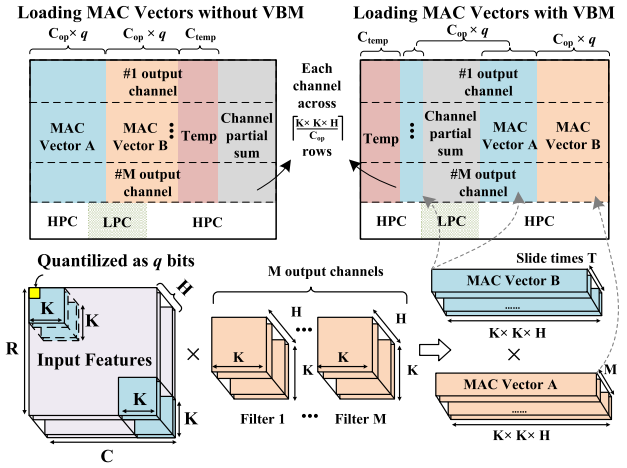


Fig. 12. Examples of loading input features and weights for in-memory MAC operations without and with mapping method.

and read operations, respectively. To summarize, Algorithm 1 can still accommodate the conventional bit-serial subtraction and division flow to map data bits into CIM array for enhancing the performance of CIM mode.

Algorithm 1 can also be employed for bit-serial floating-point arithmetic operations. The multiplication (division) of floating-point can be divided into addition (subtraction) of exponent and multiplication (division) of mantissa which are already been explained in the fixed-point mapping method. For addition and subtraction of floating-point, three steps are proposed using bit-serial processing method, namely equalizing exponent, adding mantissa, and normalizing result [4]. During the processes, each bit of two operands, temp results, and sum (or difference) are involved either in read or CIM operations. So using Algorithm 1, data for floating-point arithmetic operations can be mapped into CIM array.

C. Mapping Processes of Multiple Computation Tasks (MAC Operations) With Proposed VBM Method

In this section, the MAC operations are evaluated as examples of multiple computation tasks. The mapping process of MAC vectors is shown in Fig. 12. The size of input feature is $R \times C$ with H input channels. The filter size is $K \times K \times H \times M$, where K is the size of filter kernel and M is the output channel number. For each slide of filter, corresponding weight filters and input feature are expanded to MAC vectors A and B with size of $K \times K \times H \times M$ and $K \times K \times H \times T$, respectively, where T is the sliding times of filter. The MAC vectors A and B can be further mapped into SRAM-CIM array according to Algorithm 2. During the mapping, the array are divided into four parts, namely MAC vector A, MAC vector B, temp columns, and channel partial sum columns. The former two parts are used to store MAC vectors A and B, respectively. Due to the limited memory columns in a single row, MAC vectors of dimensions $K \times K \times H$ with q quantized bits are split to store across $\lceil (K \times K \times H \times q)/C_{op} \rceil$ rows, where C_{op} is the operand space determined in Algorithm 2. The temp columns are used to store the temporary partial results and do partial accumulations. The channel partial sum columns are used to store the final results between MAC vectors A and B in the same row. The channel partial sums in the same output channel can be stored across different rows and need to be accumulated to generate final

channel output. To support inter-row accumulations, the channel partial sums spread across different rows are read out and stored into the CIM array according to Algorithm 1 and Algorithm 2 as presented in Section IV-A until the final result is generated. In the above processes, for the case without VBM method, MAC vectors A and B can be partially mapped into the LPC regions, thus large clock cycle period is induced. For the case with VBM method, the channel partial sums are mapped into LPC regions since the channel partial sums are only involved in write operations in CIM mode, while MAC vectors and temp columns are mapped into HPC regions to enhance CIM performance.

TABLE II
PARAMETERS OF CN-MOSFETS ASSUMING 16NM TECHNOLOGY NODE

Parameters	Symbols	Value
Gate Oxide Thickness (nm)	t_{ox}	2
Gate Length (nm)	L_g	16
Contact Length (nm)	L_c	16
Extension Length (nm)	L_{ext}	16
Spacing (nm)	s	6
Gate Height (nm)	H_g	20
Flat Band Voltage for n-type CN-MOSFETs (V)	V_{fbn}	0.015
Flat Band Voltage for p-type CN-MOSFETs (V)	V_{fbp}	-0.015
Diameter (nm)	d_{UNI}	1.2

V. RESULTS AND ANALYSIS

In this section, 256 × 256-bit SRAM-CIM arrays based on different SRAM-CIM cells are evaluated and compared in HPSICE simulations, assuming 16nm carbon nanotube transistor technology with 0.7V supply voltages at 25 °C. The parameters of CN-MOSFETs of this work are listed in Table II. The parasitics of MIVs and metal interconnects of different SRAM-CIM cells are extracted by Sentaurus TCAD and included in 1000 Monte Carlo simulations. To emulate the fabrication imperfections, the diameter variations and metallic carbon nanotubes removal process are considered. The diameters of nanotube are assumed to follow the Gaussian distribution with variation of $3\sigma/\mu = 10\%$, where the nominal diameter d_{UNI} is $\mu = 1.2$nm [25]. The metallic CN probability P_m is assumed as 1%, 2%, and 3% with the probability of wrongly removing the s-CNs $P_{Rs} = 2.5\%$ [25]. The simulation results are summarized in Table III. Since the proposed M3D-4N4P CNRP and CPRN SRAM-CIM cells have symmetric structures and electrical properties, the comparison table and following analysis only include M3D-4N4P CNRP SRAM-CIM for simplicity. The analysis of layout area, data-access robustness, CIM performance, energy efficiency, and data intensive application benchmarks are discussed from Section V-A to Section V-D, respectively.

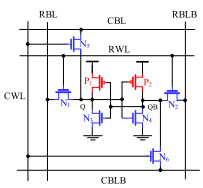
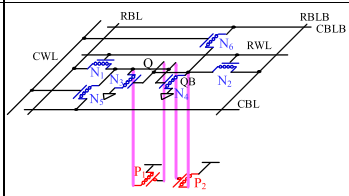
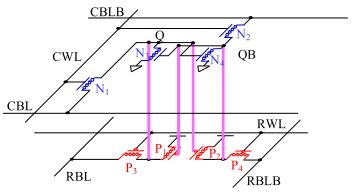
A. Layout Area

The layout areas of different SRAM-CIM cells are compared in this section, as listed in Table III. 2D-6N2P SRAM-CIM cell is designed with three metal layers for signal routing. To make the layout more compact, signals

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TABLE III

COMPARISON OF SRAM-CIM CELLS WITH 16NM CARBON NANOTUBE TRANSISTOR TECHNOLOGY ($V_{DD} = 0.7$ V, $P_m = 3\%$, $P_{Rs} = 2.5\%$)

SRAM-CIM Cell Type		2D-6N2P SRAM-CIM Cell		M3D-6N2P SRAM-CIM Cell		M3D-4N4P CNRP SRAM-CIM Cell	
							
		Without VBM	With VBM	Without VBM	With VBM	Without VBM	With VBM
Cell Area (nm^2)		134956.80		81734.40		67046.40	
Array Area (256×256-bit) (μm^2)		8844.53		5356.55		4393.95	
Functional Yield of 256×256-bit Array		98.14%		97.25%		97.98%	
RSNM (mV)	SRAM Mode	94.35		94.35		93.24	
	CIM Mode					94.35	
CWLM (mV)	SRAM Mode	173.12		173.12		152.36	
	CIM Mode					173.12	
Maximum CIM Latency (ps)		949	424	669	334	779	268
Minimum Required Clock Period for CIM (ps)		1259.61	734.61	980.41	645.41	1089.81	578.81
Maximum Allowed Clock Frequency for CIM (GHz)		0.79	1.36	1.02	1.55	0.92	1.73
Performance (GOPS) of 32-bit Fixed-Point	ADD	6.16	10.56	7.91	12.02	7.12	13.40
	SUB	3.13	5.36	4.02	6.10	3.61	6.80
	MULT	0.17	0.29	0.22	0.34	0.20	0.37
	UDIV	0.12	0.20	0.15	0.23	0.14	0.26
Energy Efficiency (GOPS/W) of 32-bit Fixed-Point	ADD	1445.65		1585.58		1860.54	
	SUB	728.54		799.73		938.28	
	MULT	45.41		49.91		58.51	
	UDIV	20.84		22.39		26.36	
Energy Efficiency Per Unit Area (TOPS/W/mm^2) of 32-bit Fixed-Point	ADD	163.45		296.01		423.43	
	SUB	82.37		149.30		213.54	
	MULT	5.13		9.32		13.31	
	UDIV	2.36		4.18		6.00	
Performance (GFLOPS) of 32-bit Floating-Point	ADD/SUB	0.04	0.07	0.05	0.08	0.05	0.09
	MULT	0.30	0.51	0.38	0.58	0.35	0.65
	DIV	0.29	0.50	0.37	0.57	0.34	0.63
Energy Efficiency (GFLOPS/W)of 32-bit Floating-Point	ADD/SUB	16.99		18.16		21.33	
	MULT	86.30		94.89		111.20	
	DIV	39.24		42.16		49.62	
Energy Efficiency Per Unit Area (TFLOPS/W/mm^2) of 32-bit Floating-Point	ADD/SUB	1.92		3.39		4.86	
	MULT	9.76		17.71		25.31	
	DIV	4.44		7.87		11.29	

shared between horizontally neighboring cells are placed on the horizontal borders of the layout. However, additional height is inevitably reserved for *CBL* and *CBLB* routing, which cannot be shared by vertically neighboring cells. The layout area of 256 × 256-bit 2D-6N2P SRAM-CIM array is estimated as $8844.53\mu m^2$ assuming 16 nm technology. To enable the transistor-level M3D design of previously published 2D-6N2P SRAM-CIM cell, 2 p-channel devices are placed on bottom layers with 4 MIVs connected with top layer where 6 n-channel devices are placed. Only one metal layer is used in bottom layer for less heat dissipation. The layout area of the M3D-6N2P SRAM-CIM array is reduced by 39.43% as compared to its 2D version. Due to balanced transistor number in top and bottom layers, less area skew is observed in the proposed M3D-4N4P SRAM-CIM array. Thus, the proposed M3D-4N4P SRAM-CIM array reduces the layout area by up to 50.32% and 17.97% compared to previously published 2D and M3D-6N2P SRAM-CIM arrays, respectively.

B. Evaluations of Data-Access Robustness

The data-access robustness of different SRAM-CIM cells are compared in this section, as listed in Table III. 2D and M3D- 6N2P SRAM-CIM cells have the same read margin due to the same read-path ratios in SRAM and CIM modes, which is simulated as 94.35mV. For the similar reason, 2D and M3D-6N2P SRAM-CIM cells have the same write margin in SRAM and CIM modes, which is simulated as 173.12mV. Due to the reduced read-path and write-path ratios, the proposed M3D-4N4P SRAM-CIM cell experiences small degradations in read and write margins in SRAM mode compared to the previous 2D and M3D-6N2P SRAM-CIM cells. Nevertheless, the read and write margins of proposed M3D-4N4P SRAM-CIM cell

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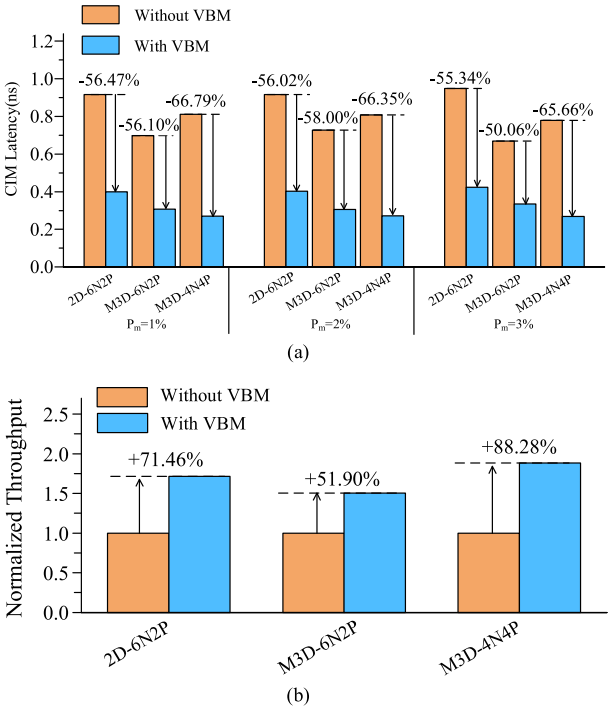


Fig. 13. (a) CIM delay simulations with different SRAM-CIM cells considering different P_m variations and fixed $P_{Rs} = 2.5\%$. The simulations are done with 256×256-bit SRAM-CIM arrays. (b) Evaluations of throughput of arithmetic operations considering $P_m = 3\%$ with fixed $P_{Rs} = 2.5\%$.

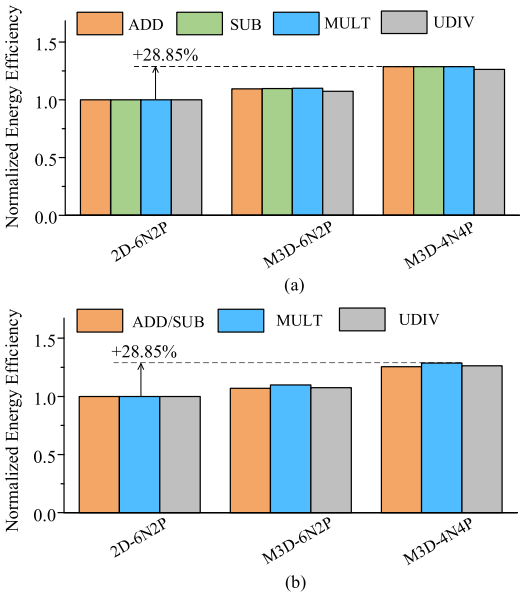


Fig. 14. Normalized energy efficiency evaluations of 2D-6N2P, M3D-6N2P, and M3D-4N4P SRAM-CIM arrays on (a) fixed-point and (b) floating-point arithmetic operations.

are still sufficiently high in both SRAM and CIM modes, which guarantees the robustness of data-access.

C. Evaluations of CIM Performance and Energy Efficiency

The CIM performance of different SRAM-CIM arrays are evaluated in this section. The evaluations of basic logic operations are shown in Fig. 13(a). The process variations of CN-MOSFETs including diameter variations with metallic removal process are considered with spatial correlations. For the case without VBM method, 2D-6N2P SRAM-CIM array has the longest CIM latency because of the largest parasitic effect on *CBL* and *CBLB* caused by the large footprint as compared to the other two M3D CIM arrays. Though the proposed M3D-4N4P SRAM-CIM array has a more compact layout with smaller parasitic effect, it shows worse performance compared to M3D-6N2P SRAM-CIM array in the case without VBM method. This is because the discharge paths of proposed M3D-4N4P CNRP SRAM-CIM cell in CIM operations are more likely to be extremely slow as compared to the previously published 6N2P SRAM-CIM cells due to the spatial correlations between pull down transistors (N_3 and N_4) and access transistors (N_1 and N_2), while M3D-6N2P SRAM-CIM cell has relatively small parasitic effect and no correlated discharge paths. Therefore, M3D-6N2P SRAM-CIM array has the shortest latency among the three SRAM-CIM arrays when the mapping method is not considered. For the case with VBM method, high-performance SRAM-CIM cells are used in CIM operations, resulting in performance improvements in all the three SRAM-CIM arrays. Because the three types of SRAM-CIM cells are designed with the same strength of discharge path, the difference of CIM latency are strongly related to the parasitic effects. Therefore, the proposed M3D-4N4P SRAM-CIM array has the best performance out of three SRAM-CIM arrays due to the most compact layout. Overall, the CIM latency of proposed M3D-4N4P SRAM-CIM array is reduced by up to 65.66% compared to the case without VBM method.

The throughput of arithmetic computing of different types of SRAM-CIM arrays are shown in Fig. 13(b). With VBM method, the dominated CIM latency in cycle period is reduced in three types of SRAM-CIM arrays. As compared to the case without VBM method, the 2D-6N2P, M3D-6N2P, and M3D-4N4P SRAM-CIM arrays with VBM method achieve 71.49%, 51.90%, and 88.28% higher throughput for the arithmetic computing, respectively. The proposed M3D-4N4P SRAM-CIM array with the most compact layout achieves the highest throughput among all the designs, demonstrating up to 2.17× higher throughput on arithmetic computing compared to 2D-6N2P SRAM-CIM array without mapping.

The energy efficiency evaluations of fixed- and floating-point arithmetic operations for different SRAM-CIM arrays are listed in Table III. The normalized energy efficiency results are shown in Fig. 14. To reduce the power consumptions of bitlines, the pulse widths of wordlines in different SRAM-CIM arrays are individually tuned to reduce the voltage swing of bitlines. Furthermore, with ultra-high dense layout which incurs less parasitic effect with less power consumptions, the proposed M3D-4N4P SRAM-CIM array achieves up to 28.85% higher energy efficiency compared to 2D-6N2P SRAM-CIM array in both fixed- and floating-point multiplications, as shown in Figs. 14(a) and (b). A compound index named energy efficiency per unit area is also proposed to illustrate the superiority of energy efficiency and density of different SRAM-CIM cells. The proposed M3D-4N4P SRAM-CIM array achieves 2.52× higher energy efficiency per unit area than 2D-6N2P SRAM-CIM array.

D. Evaluations of MAC-Based Benchmarks

The evaluations of MAC-dominated applications are shown in Fig. 15. Data intensive benchmarks like Cuda-Convnet, Alex FC, and FIR are evaluated with different SRAM-CIM

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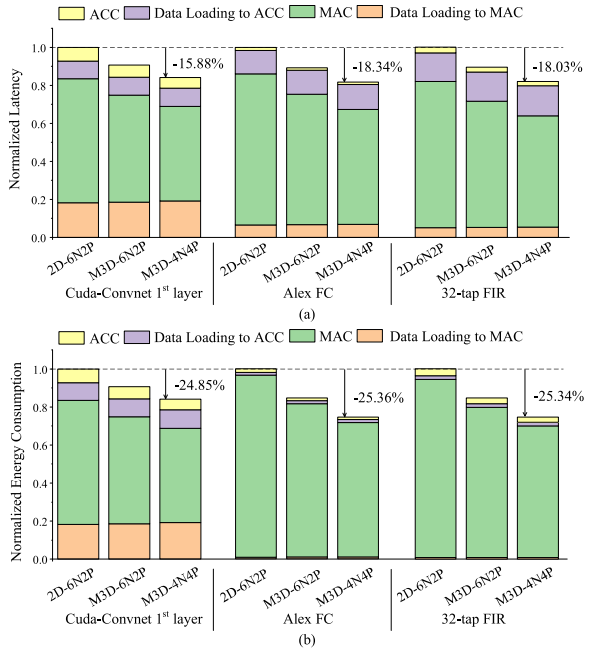


Fig. 15. Evaluations of 2D-6N2P, M3D-6N2P, and M3D-4N4P SRAM-CIM arrays with mapping method with P_m equal to 3% on Cuda-Convnet 1st layer, AlexNet last layer, and 512 32-tap FIR. (a) Latency evaluations. (b) Energy evaluations.

arrays considering VBM method. The three benchmarks are mapped into SRAM-CIM arrays using the Algorithm 1 and Algorithm 2 introduced in Section V-A. Two types of SRAM-CIM arrays for 2D-6N2P, M3D-6N2P, and M3D-4N4P are used in the evaluations, namely MAC array and accumulation array (ACC array). The MAC array is used to generate channel partial results, and the ACC array is used to further reduce channel partial results to final results. To enable the applications, both SRAM and CIM modes are used for data loading and CIM, respectively. Due to the limited memory columns, long MAC vectors needs to be split into several rows. The partial results produced by the partial MAC vectors in each row will be accumulated in the ACC array. Among all the designs, the proposed M3D-4N4P SRAM-CIM array achieves the best performance and energy efficiency thanks to its compact layout design, with a maximum 18.34% performance gain and up to 25.36% reduced energy consumption in evaluations of Alex FC compared to the previously published 2D-6N2P SRAM-CIM array.

VI. Conclusion

New bit-serial M3D-4N4P SRAM-CIM cells based on carbon nanotube transistor technology are proposed in this paper for realizing high-performance and robust in-memory computing. The profiles of SRAM-CIM cells are optimized with quality metric based sizing method by comprehensively considering memory metrics for reliable data-access. Furthermore, the proposed variation-aware bit-wise data-mapping method using the high-performance cell in SRAM-CIM significantly enhances the performance despite of the specific process non-idealities in CN-MOSFETs. Finally, the areas of proposed bit-serial M3D-4N4P SRAM-CIM cells are reduced by 50.32% and 17.97% as compared to the previously published bit-serial 2D and M3D-6N2P SRAM-CIM cells assuming CN-MOSFETs, respectively. The proposed M3D-4N4P SRAM-CIM array also achieves by up to 2.17× higher throughput on arithmetic computing task and 18.34% lower computing latency with 25.36% reduced energy consumptions on MAC-intensive benchmarks, respectively, compared to the previous 2D-6N2P SRAM-CIM array.

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